Olist E-Commerce Analysis

SQL and Tableau analysis of the Brazilian Olist marketplace with business recommendations.

Greg Lauray | April 2026

Project Overview

This project analyzes the Brazilian E-Commerce Public Dataset by Olist to surface actionable insights for Olist's marketplace team. Using a normalized MySQL database and Tableau dashboards, the analysis addresses three core business questions:

1. Which product categories are underperforming on customer reviews, and what are the next steps to address them?
2. How does demand for seasonal categories (air conditioning) trend over time, and how should Olist prepare for peak season?
3. Which Brazilian states drive the most customer volume, and what does performance look like in those key markets?

Dataset

- **Source:** Brazilian E-Commerce Public Dataset by Olist (Kaggle)
- **Time range:** September 2016 – October 2018
- **Scale:** ~96,000 orders across 8 related tables (customers, orders, order items, products, sellers, payments, reviews, and category translations)

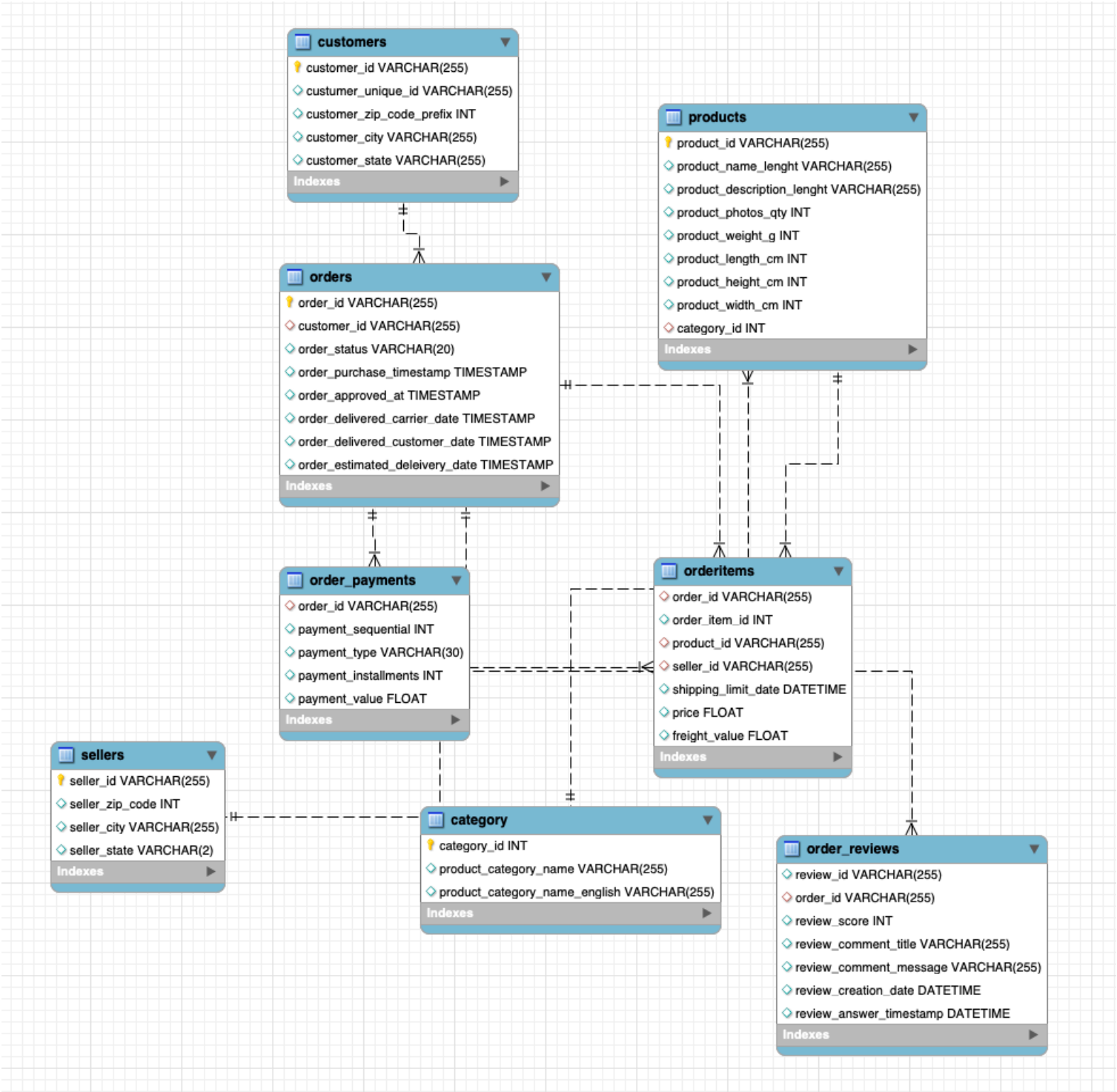
Tools & Techniques

Category	Tools
Database	MySQL Workbench
Visualization	Tableau Public
Version Control	GitHub
SQL Techniques	CTEs, joins, window functions (DENSE_RANK with PARTITION BY), conditional aggregation

Data Model

The raw dataset was provided as 9 CSV files. I designed a normalized MySQL schema with 8 tables, primary and foreign key constraints, and a separate category lookup table to eliminate string duplication in the products table.

ERD:


customers
customer_id VARCHAR(255)
custumer_unique_id VARCHAR(255)
customer_zip_code_prefix INT
customer_city VARCHAR(255)
customer_state VARCHAR(255)
Indexes
products
product_id VARCHAR(255)
product_name_lenght VARCHAR(255)
product_description_lenght VARCHAR(255)
product_photos_qty INT
product_weight_g INT
product_length_cm INT
product_height_cm INT
product_width_cm INT
category_id INT
Indexes
orders
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customer_id VARCHAR(255)
order_status VARCHAR(20)
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order_approved_at TIMESTAMP
order_delivered_carrier_date TIMESTAMP
order_delivered_customer_date TIMESTAMP
order_estimated_deleivery_date TIMESTAMP
Indexes
order_payments
order_id VARCHAR(255)
payment_sequential INT
payment_type VARCHAR(30)
payment_installments INT
payment_value FLOAT
Indexes
orderitems
order_id VARCHAR(255)
order_item_id INT
product_id VARCHAR(255)
seller_id VARCHAR(255)
shipping_limit_date DATETIME
price FLOAT
freight_value FLOAT
Indexes
sellers
seller_id VARCHAR(255)
seller_zip_code INT
seller_city VARCHAR(255)
seller_state VARCHAR(2)
Indexes
category
category_id INT
product_category_name VARCHAR(255)
product_category_name_english VARCHAR(255)
Indexes
order_reviews
review_id VARCHAR(255)
order_id VARCHAR(255)
review_score INT
review_comment_title VARCHAR(255)
review_comment_message VARCHAR(255)
review_creation_date DATETIME
review_answer_timestamp DATETIME
Indexes

Dashboard

Tableau Public Dashboard: [click here for tableau public]

Key Findings

1. São Paulo dominates customer volume

São Paulo accounts for ~54.4% of all Olist customers, far outpacing Rio de Janeiro (14%) and Minas Gerais (12%).

2. Air conditioning revenue is sharply seasonal

Air conditioning revenue surged 347% from its August 2017 trough ($199.70) to its November 2017 peak ($891.80), before contracting 66% into December. The pattern reflects Brazil's summer onset combined with Black Friday concentration.

3. Five categories show chronically low review scores

alimentos, cama mesa banho, fraldas higiene, móveis decoracao, and telefonia all carry average review scores below 3.0, with at least 10 reviews each.

4. São Paulo's top sellers maintain consistently high review scores

The top 5 sellers in SP show average review scores ranging from 4.75 to 4.96, indicating strong customer satisfaction across leading sellers in the region.

5. Platform-wide performance baseline

Total revenue of ~$1.3M across 8,981 delivered orders, with an average order value of $145 and an average review score of 4.0, indicating broadly positive customer satisfaction.

6. On-time delivery rates vary significantly by state

Northern states (AC, AM, AP) achieve near-100% on-time delivery, while AL and other northeastern states show the weakest performance — a regional logistics signal worth investigating.

7. Top-performing categories in São Paulo cities

automotivo and esporte_lazer appear among the most-purchased categories across São Paulo cities, providing category-level targeting data for regional marketing.

Recommendations

1. Apply NLP to diagnose low-review categories

The products flagged in the review analysis share low scores but the underlying causes are unknown — reviews could reflect product quality issues, delivery problems, seller communication failures, or mismatched product descriptions. Applying topic modeling (LDA or BERTopic) and sentiment analysis to the review_comment_message field for these specific products would categorize complaints into actionable buckets, converting the current "which products to investigate" list into a prioritized action plan by failure mode.

2. Align air conditioning marketing and inventory with the Q4 demand curve

With air conditioning revenue climbing 347% from August to November, Olist and its sellers should launch targeted marketing campaigns in September — ahead of the seasonal ramp — rather than reacting to November demand. Inventory and seller capacity should be staged in October to meet the Black Friday surge and avoid stockouts, then tapered in December to prevent overhang into a low-volume January.

Limitations & Future Work

Limitations

- The Olist dataset ends in October 2018, limiting full-year 2018 analysis and cross-year Q4 comparison.
- 2016 data is sparse (the platform launched in September 2016), so early-year trends reflect platform launch dynamics rather than steady-state behavior.

- Review comments are in Brazilian Portuguese, which would require a Portuguese-language NLP model (e.g., BERTimbau) rather than default English models.
- Written review coverage is incomplete — many reviews have a numeric score but no textual comment, limiting the reach of any sentiment analysis.

Future Work

- NLP analysis on Portuguese review comments to categorize dissatisfaction drivers.
- Seller-level deep dive on logistics performance and regional delivery patterns.

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Skills:

SQL · Python · scikit-learn · FastAPI · Pydantic · SQLAlchemy · Tableau